How Does AI Work?

Part 1: Pattern Recognition

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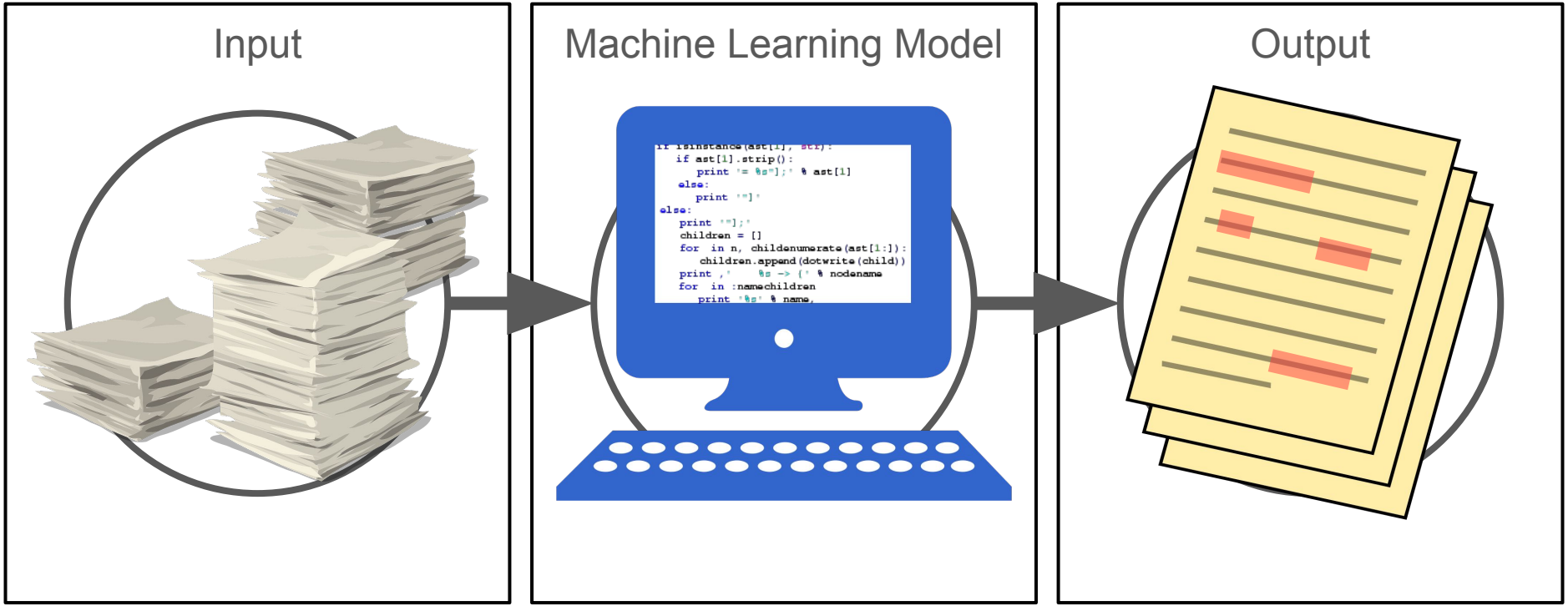
Overview of Steps



Input

- The input is data that has been "pre-processed", or prepared into a format that AI can use, usually by humans using a simple program
  - Text is converted into a string of characters or words
  - Images are converted into arrays of pixels
    - In the picture, pixel colors are converted into three values representing red, green, and blue (RGB).

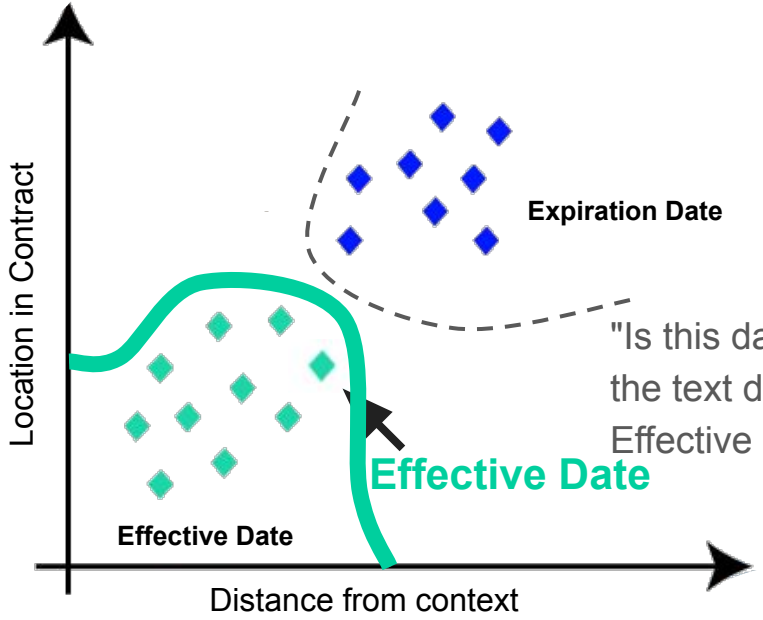
Source: https://blogs.mathworks.com/steve/2012/12/04/image-effects-part-4/

Machine Learning Model

The model utilizes
Pattern Recognition
to predict and classify data inputs.

Pattern Recognition

- Recognize patterns in data
- Group and classify data points based on features
- One type of simple pattern recognition is "Nearest Neighbor"



"Is this date in the contract more close to the text describing an Expiration Date or Effective Date?"

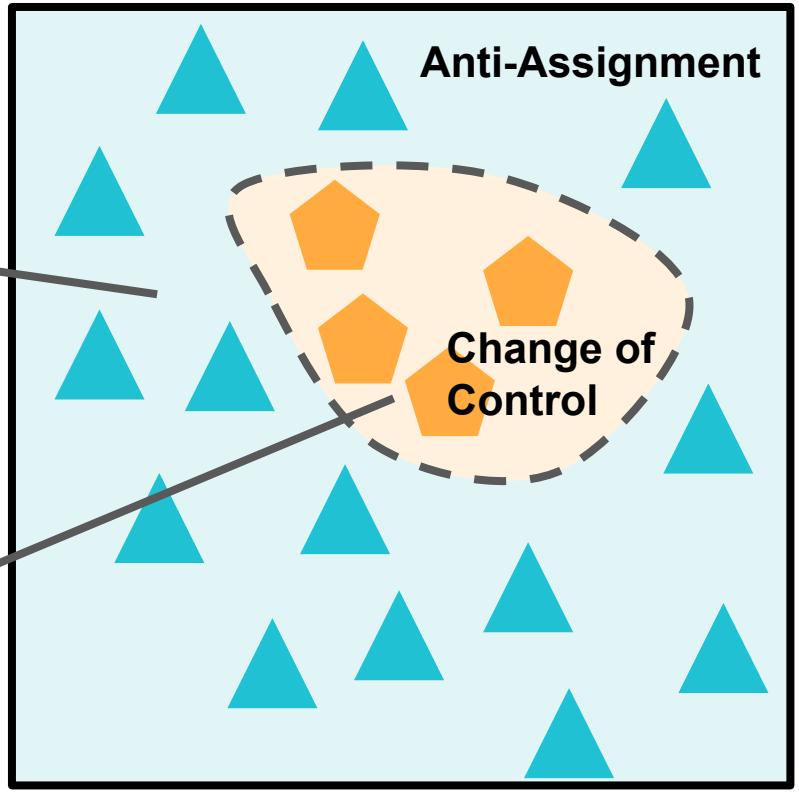
The Nearest Neighbor search looks through the existing data points for the closest one to the new data point.

Use Cases:

- Plagiarism detection
- Spell-check

Pattern Recognition (cont.)

- Example:
  Classification based on contract clauses

An Anti-Assignment clause means the contract can't be assigned without consent or notice.

A Change of Control clause is a type of Anti-Assignment clause where **assignment would include a change in control of a contracting party.**